



SECI 1113: COMPUTATIONAL MATHEMATICS
20232024 – SEMESTER 2

ASSIGNMENT #4

(Submission Date: 23rd June. 2024)

1. Given a matrix A as follows.

$$A = \begin{bmatrix} 0 & 11 & -5 \\ -2 & 17 & -7 \\ -4 & 26 & -10 \end{bmatrix}$$

- a) Find the characteristics of polynomial of the matrix A . (5 marks)
- b) Find the dominant eigenvalue, λ_1 and the corresponding eigenvector of matrix A using **Power method**. Use $v^{(0)} = [0, 1, 1]$. Do calculation in 3 decimal points and take $\varepsilon = 0.005$. Round your eigenvalue, λ_1 to the nearest integer. (5 marks)
- c) Use result in (b) to find the smallest eigenvalue, λ_3 and the corresponding eigenvector using **Shifted Power method**. Use $v^{(0)} = [1.0, 0.25, 1.0]$. Do calculation in 3 decimal points and take $\varepsilon = 0.005$. Round your smallest eigenvalue, λ_3 to the nearest integer. (8 marks)
2. The viscosity μ of the fluid depends on its temperature T according to the relationship represented by the data in Table 1.

Table 1

$T (^{\circ}\text{C})$	5	20	30	50	55
$\mu (\text{N-sec/m}^2)$	0.0800	0.0150	0.0090	0.0060	0.0055

Use the following methods to estimate for the viscosity at $T = 40$.

- i. Appropriate Newton's interpolation (5 marks)
Do calculation in 6 decimal points.
- ii. Lagrange interpolation. (5 marks)

3. Company X somehow believes the amount paid per hour to the labor has relation to the profit obtain by the company. Table 2 shows the profit and the amount pay to the labor for several months.

Table 2

Profit	94	118	147	180	230
Labour cost (S)	10	16	25	40	60

- a) Find the least square values for the following constants in
- $$\text{Profit} = a + bS \quad (10 \text{ marks})$$
- b) Find the profit if the labour cost is 210. (2 marks)
4. Given, $f(x) = 2x^3 + x^2 - 4$, estimate $f'(2.5)$ and $f''(2.5)$ using
- a) Central three-point and five-point formulas with $h = 0.5$.
- b) Find the exact value of $f'(2.5)$ and $f''(2.5)$.
- c) Calculate the error for each estimation in 4(a).
5. Approximate $\int_0^3 \frac{2}{1+5x^2} dx$ using:
- a) Trapezoidal rule with $N=6$.
- b) Simpson's 1/3 formula with $N=6$.
- c) Simpson's 3/8 formula with $h=0.25$.
- d) Romberg integration. Use $\varepsilon = 0.005$

Note:

1. Do calculation in 4 decimal points unless stated otherwise.
2. You need to solve them using manual calculations and develop the Python code to verify the answer. Both solutions must be submitted.
3. For the Python code, write it in one file with a proper title.
4. The solution using Python contributes 30 marks.